



The Value and Application of Car Head-Up Display in Interactive Design

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ABSTRACT

The HUD (head-up display) technology was first applied to aircraft. The collected data is displayed on the windshield panel of the aircraft. The purpose is to avoid errors when the pilot looks down at the required data, resulting in some irreversible problems. as a result of. The so-called head-up display refers to the integration of the flight assistance technology of the aircraft into the car. The automobile HUD also integrates some important driving data and projects it on the front windshield, and feeds it back to the driver, reducing the frequency of looking down at the instrument, preventing the driver from distracting, and reducing the occurrence of traffic accidents. The purpose of this research is to combine HUD technology with interactive design to improve the safety of vehicle driving.

This research takes as the starting point which functions should be provided by the automobile HUD, and provides a reasonable “export” between the conversion between aircraft and automobile. Perfect integration of traditional cars and modern technology, thereby reducing the occurrence of car accidents. At the same time, the user-centered design ideal is used to enable consumers to quickly accept and proficiently use the technology.

CCS CONCEPTS

• Human-centered computing~Interaction design~Empirical studies in interaction design

KEYWORDS

Panoramic, feedback, Fusion, Interactive design, Head-up display

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1 Introduction

Traffic accidents caused by automobiles happen every day, and the number of accidents, the damage caused, or the number of casualties are shocking. This research uses the combination of HUD (head-up display) technology and interactive design to reduce car traffic accidents caused by vision problems.

With the development of the times, people are increasingly pursuing products with individuality, independence and a sense of technology. As the forefront of science and technology, airplanes are unparalleled in every technology about it. What is impressive is the feedback of various data on the front windshield of the airplane. The windshield is to reduce the visual field error caused by the driver between raising and lowering his head, and to avoid flight accidents [1]. And why can't cars with the same windshield have this technology? If this technology is used, will traffic accidents caused by cars be reduced accordingly?

At present, a large number of on-board equipment and on-board software are constantly pouring into the car [2]. It has become a normal state for drivers to interact with these software and devices. They check or send information while driving, look up addresses or make calls on the on-board system, and are in the car. Various common behaviors, such as selecting music or listening to channels of interest on the Internet, not only bring convenience to the driver, but also increase the risk of driving.

Car ownership continues to increase, and people spend more and more time in cars [3]. Automobiles should become the forefront of current technology, digitization, and networking, which also provides a good stage for interaction design. Based on this research, a panoramic HUD product specially designed for automobiles was designed.

2 Development Status of Automotive HUD

2.1 HUD Product Status

The use of HUD technology in the car is to ensure that the driver can see the information he wants to see as much as possible without lowering his head when driving safely and stably [4], thereby reducing the visual fatigue caused by the frequent switching of the driver's far and near perspective, thereby reducing traffic The accident happened.

At present, automotive HUD products are mainly divided into three categories: automotive manufacturers pre-installed HUD, automotive aftermarket HUD equipment and mobile phone HUD application projection.

Although automotive HUD products continue to develop, the current pre-installed HUD by car manufacturers only appear on a small number of high-end brand models, such as BMW 5 Series Li, Audi A6/A7 and Lexus GS. Land Rover has also upgraded its HUD projection technology on some models and promoted it as a selling point.

Most HUD products in the market are automotive aftermarket HUD [5]. A smaller HUD product is installed in front of the driver. This product is more convenient to install and disassemble and has a high degree of freedom. But there are also many drawbacks, such as the inability to match cars well, the inability to obtain higher authority for the car, and less information that can be displayed.

2.2 HUD Product Analysis

Carrobot (as shown in Fig. 1) is a HUD product in the automotive aftermarket. It displays information through the HUD transparent projection screen placed on the dashboard directly in front of the steering wheel. It can navigate, make calls, send and receive text messages, and listen through voice control. Songs and other functions. Allow drivers to focus on driving while safely taking into account the needs of navigation, communication, entertainment, and social interaction, making driving safer and more enjoyable. The advantage is that it is easy to install; the disadvantage is that the style is single, not beautiful enough, the display range is small, and the display information is single.

There are also some HUD that are projected onto the front windshield of the car through a mobile phone application. These products use the mobile phone as the on-board HUD of the mobile terminal. Such as HUDMAP and HUDWAY (as shown in Fig. 2) and so on. A serious problem with mobile HUD is light. The HUD feedback effect is better when the light is dim in underground garages, tunnels, etc., but when the car enters the outdoors, the light is strong at this time, and the display effect is not ideal. It is difficult for the driver to see the information displayed by the projection, even with the mobile phone. Turning up the screen brightness doesn't help. And it is extremely susceptible to interference from ambient light penetrating from the outside.

The advantages of pre-installed HUD by car manufacturers are: exclusive design, high product fit, distinctive brand characteristics, more mature imaging technology, and will not occupy more space

in the car [6]. In addition, the linkage between the pre-installed HUD and other systems in the car is stronger, and it will have a great advantage in the future development of HUD.



Figure 1: Automotive aftermarket HUD - Carrobot



Figure 2: HUDWAY HUD mode usage scenarios

3 HUD Design Principles

3.1 Design Positioning

With the development of the times, both electronic technology and the Internet have made qualitative breakthroughs. People are increasingly inseparable from the convenience brought by the Internet. Through the integration of information, the application of the Internet of Things has become more and more extensive [7]. For example, smart watches, smart water and electricity meters, and even shared bicycles. However, the progress of the automotive industry is not so obvious.

The slow development of automobiles has its own reasons. The development of the automobile industry is usually slower than other products of the same generation. This is due to objective reasons such as long automobile replacement cycles and high costs. According to market research, the frequency of mobile phone replacement is 2-4 times a year, while automobiles are basically replaced every 5 years. This leads to the fact that even

the latest electronic products used in automobiles are scientific and technological achievements 5 years ago.

Nowadays, automobiles have been fully integrated into people's lives [8]. Therefore, consumers have higher and higher expectations for automobile products and are increasingly interested in automobile product innovation. This research takes the car as the carrier and realizes through a series of functions to let people feel the change of the car. At the same time, the user-centered design concept is used to bring consumers a different interactive experience through innovative interactive design methods. Let people and cars be harmonious and unified, making cars more in line with contemporary technological trends and characteristics of the times.

3.2 Design Goals

The interaction of the car is mainly reflected in the instrument panel and the center console [9], and the focus of this research is also mainly concentrated on this. One of the development directions of the automobile interactive interface is the head-up display. The head-up display directly projects the information that needs to be displayed on the front windshield panel to generate virtual graphics.

The head-up display cannot replace the traditional instrument combination, but has advantages in the visual experience of safety information and interface interaction. Especially the combination of head-up display and augmented reality technology can create a good user experience.

This practical project pre-built a panoramic HUD, so the information feedback of the front windshield should not be limited to the display of one or two information. The design goal of this project is to centrally feedback the information needed by the driver to a specific location. This information should first include the basic functions of the center console, and the road conditions and vehicle information ahead and the information for the driver's entertainment should also be displayed.

Since the driver needs to know more vehicle information when driving, the content display volume is large, and the position and size for information feedback are limited by the front windshield, so it is necessary to make reasonable choices about the display content. This requires that the feedback of each information must be rigorous and independent, and too much and unimportant information cannot be displayed, which will cause safety hazards to the car.

A moderate sense of technology can improve the driver's user experience [10], but excessive information display can also have a negative impact on the user experience. From the perspective of the driver, the information integration of the car's interactive interface should first consider that the driver's line of sight cannot deviate from the road ahead when the vehicle is driving. During the driving of the car, all information should be concentrated on the front windshield for display, and the driver's operations should also be fed back on the front windshield as much as possible. Appropriate information should appear in the right scene to avoid excessive and useless the message appears. For example, when there is an incoming call, the first thing to display is the

caller information. For another example, when listening to music, you only need to provide information about volume adjustment and song title when driving, without displaying the lyrics.

Second, consider where the information is displayed. Information related to assisted driving is important data in the driving process of the vehicle, and should be placed in the main field of view of the display direction, such as navigation information, reminder of excessive vehicle speed, reminder of fuel level, etc. Information not related to driving, such as the current date, song information, etc., can be placed in a secondary location for display, so as to provide the driver with a large enough vision so that the driver can cope with complex road conditions at any time.

4 Application Design of Panoramic HUD

4.1 Personalization and Customization

Personalization and customization can significantly improve the user experience. The key factor of personalization is how to better integrate identification technology into in-vehicle applications and services. The development of recognition technology is an important factor driving personalization and customization.

At present, automobile personalization and customization are mainly parametric customization. From a design point of view, personalization and customization should be the personalization and customization of users in the process of use. The panoramic HUD of this project concentrates feedback on the front windshield of the information needed during the driving of the vehicle. In order to distinguish personas, identification technology can distinguish different users, authorize different types of users to use different devices, and perform personalized switching according to the set user mode.

Biometrics technology is developing rapidly, and currently mature solutions include facial recognition, joint recognition, signature recognition, red membrane recognition, and retinal scanning. However, considering the actual use of the car [11], fingerprint recognition (as shown in Fig. 3 and Fig. 4) and voice recognition (as shown in Fig. 5) technologies are selected for this project.



Figure 3: Fingerprint recognition technology used in cars



Figure 4: Fingerprint recognition technology used in cars



Figure 5: Speech recognition technology used in cars

4.2 "Ubiquitous" Display

"Ubiquitous" here refers to the entire front windshield. Every place can be used to display data, but it needs to be typeset according to the primary and secondary relationship.

From the perspective of the display mode of the car, the HUD will gradually assume part of the display function of the dashboard to reduce the traffic accidents caused by the driver's eyes moving away from the road ahead due to the lowering of the head. The display content of HUD should be based on driving information related to assisted driving, such as vehicle speed and navigation. On this basis, a small amount of non-important information, such as entertainment systems, can also be displayed on the HUD, but it should be in a relatively secondary position, or displayed and hidden according to the driver's needs and operations. This can further enhance personalization and improve user experience.

Panoramic HUD integrates the information and feeds it back to the front windshield. The original driving-centric information architecture will be decomposed in various levels such as driving, interaction, and entertainment. The displayed carrier and location are closely related to the information interactors. Related. For example, the driver can manipulate the relevant information used by the vehicle, and other passengers in the vehicle can participate

in the interaction of social and entertainment information to enhance the user experience.

4.3 Gesture Interaction

Gesture interaction is a new way of interaction using gesture recognition technology [12]. This project is based on the HUD interactive design in the automotive environment, mainly using gestures to interactively operate navigation, music, etc.

Gesture interaction technology can track the driver's gestures, so that the driver can complete a variety of operations without touching the physical buttons on the car while driving. Its use scene is similar to somatosensory games, and the driver can be in a specific area of the car Make specific gestures to control the car system (as shown in Fig. 6 and Fig. 7).



Figure 6: Gesture recognition technology used in cars

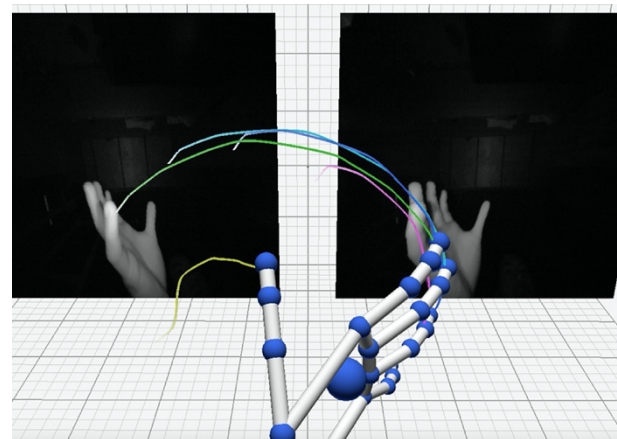


Figure 7: In the project, the use of LEAP MONTION to simulate car gesture recognition research

Gesture interaction aims to meet certain needs of the driver with simpler actions, so that the driver will not be distracted by

operating the physical buttons during driving. The driver only needs to place his palm on a certain area to complete a certain operation, and once the palm is retracted, he can exit the current operation.

4.4 Augmented Reality

The combination of augmented reality technology and automobile HUD helps to make up for the lack of natural display in the current car driving, and helps to enhance the car driving experience and sense of technology. This project uses HUD and augmented reality technology to display information such as navigation, obstacle detection, tire pressure monitoring, pedestrian detection, curve reminders, driving data, hazard warnings, etc.

For example, during the day and night, the visual range of people is very different. When driving at night, the illumination range of the low beam is small. At this time, it needs to be detected by the on-board radar. The detection range of the radar is larger than the illumination range of the low beam. The HUD can remind the driver to pay attention to pedestrians or pedestrians in the dark outside the field of view. Road barriers in blind spots. There may be some sharp turns when driving on mountain roads, and the curve description can be displayed in these areas to improve driving safety.

4.5 Enhanced Mobile Devices

At present, people are inseparable from the convenience brought by mobile phones, so it has important practical significance to combine mobile phones with cars. Cars cannot replace all the functions of mobile phones. The most effective way to integrate the two is to synchronize and adapt the two, such as Apple's Carplay, but it is limited to the interaction after the vehicle is started. The advantage of connecting the car-machine system with the mobile phone is that the mobile phone can be used to remotely control the car, which can replace certain central control functions, and can carry out special functions such as "one-key car search" and "remote start air conditioning".

4.6 Switching Between Driving and Parking Modes

The panoramic HUD is expected to replace some of the functions of the vehicle's central control system. Therefore, the dual system switching of driving and parking modes is particularly important. When the car is stopped or parked, more feedback information can be displayed on the front windshield, such as the self-check information of the car. When the vehicle is not started, the panoramic HUD can be used to watch videos and even play games, because the driver does not need to pay attention to the surrounding environment. The operation in this mode can be understood as moving the central control system to the front windshield panel. Although the driver's front view is partially blocked, the vehicle does not start or drive and can be temporarily avoided. This kind of complex operation should be shielded during the driving of the car, and only the driver is allowed to perform simple manipulations, such as gesture operation and

voice control. At this time, the HUD can only see the basic information of the feedback, so as to avoid traffic when the user performs complex operations accident.

When the car is running on different road conditions, the driver can also choose different operating modes. When the car is driving on the highway, the speed is relatively fast at this time, and too much information cannot be displayed to avoid obstructing the driver's line of sight or causing the driver to be distracted. At this time, the information fed back by the HUD is mainly based on the distance between the vehicles, the road prompts, and the navigation prompts. When a car is driving at a low speed on a city road, HUD will focus on feeding back information between the car and the person, such as surrounding people or vehicles, and reminders of violations.

5 Conclusion

This practical project is developed around the interactive design of automobile HUD. Through competitive product analysis and user research, the design process, function point analysis and content requirements were determined, and the user experience design of the panoramic HUD was carried out in accordance with the user-centric design concept.

Currently, HUD has entered a stage of rapid development, and the development of HUD is inseparable from the intervention of interaction design. Interaction design can't just stay in the original research field. It should keep up with the trend of the times, discover hot issues in the industry in time, and solve user pain points. Combining design and technology can generate more sparks. It is expected that interactive design can provide assistance for the development of HUD.

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